22AIE215 Introduction to Python
Assignment 1

Date: 10/09/2023 **Due Date: 22/9/2023**

■ Course Outcomes

- CO1: Solve problems using Python conditionals and loops.
- CO2: Apply Python functions and function calls to solve problems.
- CO3: Apply Python data structures to represent complex data.
- CO4: Develop Python Packages for reusability

- Answer all Questions

<table>
<tr><th>S. No</th><th>Questions</th><th>Marks</th><th>CO</th><th>BTL</th></tr>
<tr><td>1</td><td>Implement a program that generates a calendar for a given month and year.</td><td>5</td><td>1</td><td>1,2,3</td></tr>
<tr><td>3</td><td>Create a function in Python that accepts two parameters (both are numbers). The first will be the month as a number, and the second will be the four-digit year. The function should parse the parameters and return True if the month contains a Friday the 13th and False if it doesn't.</td><td>5</td><td>1,2</td><td>1,2,3</td></tr>
<tr><td>4</td><td>Create a function in Python that accepts one parameter: a string that's a sentence. This function should return True if any word in that sentence contains duplicate letters and False if not.</td><td>5</td><td>1,2</td><td>1,2,3</td></tr>
<tr><td>5</td><td>Create a new dictionary database called myClass as an empty dictionary. Add the following key-value pairs:
<table>
<tr><th>Keys</th><th>Values</th><th>Types</th></tr>
<tr><td>"Instructor"</td><td>"Anna"</td><td>String</td></tr>
<tr><td>"Assistants"</td><td>["Ted", "Nat", "Wei Meng"]</td><td>List</td></tr>
<tr><td>"Students"</td><td>20</td><td>Int</td></tr>
</table>
i. Add a fourth assistant to your database, named Mark
ii. Someone from another class has joined this class. add the students count by 1
iii. Print out everything you know about your database.
Expected Output:
{"Instructor": "Anna", "Assistants": ["Ted", "Nat", "Wei Meng", "Mark"], "Students": 21 }</td><td>5</td><td>1,2</td><td>1,2,3</td></tr>
<tr><td>6</td><td>Create a function that takes a string S. The task is to print all unique permutations of the given string in lexicographically sorted order.
Example 1:</td><td>5</td><td>1,2</td><td>1,2,3</td></tr>
</table>

	Input: ABC Output: ABC ACB BAC BCA CAB CBA Explanation: Given string ABC has permutations in 6 forms as ABC, ACB, BAC, BCA, CAB and CBA.			
7	Create a function that takes two strings and find the length of the longest common substring.	5	1,2	1,2,3
8.	Implement a function to calculate the greatest common divisor (GCD) of two numbers using the Euclidean algorithm.	5	1,2	1,2,3
9.	Create a program that reads a paragraph (150 words long) and counts the occurrences of each word, ignoring punctuation and case.	5	1,2	1,2,3
10.	Write a function that converts a decimal number to its binary representation.	5	1,2	1,2,3
11.	Create a program that generates a random password with a specified length using a combination of letters, digits, and symbols.	5	1,2	1,2,3
12.	Write a function that calculates the Hamming distance between two strings of equal length.	5	1,2	1,2,3
13.	Implement a program that generates a calendar for a given month and year.	5	1,2	1,2,3
14.	Write a function to check if a given number, is a perfect square.	5	1,2	1,2,3
15.	Create a program that takes a list of names and sorts them alphabetically by last name	5	1,2	1,2,3
16.	Write a function that calculates the median of a list of numbers.	5	1,2	1,2,3
17.	Write a function that finds the longest palindrome substring within a given string.	5	1,2	1,2,3
18.	Implement a program that converts a Roman numeral to an integer.	5	1,2	1,2,3
	Print the following pattern using nested loops: 54321 5432 543 54 5	5	1,2	1,2,3
19.	Print a diamond pattern with a given number of rows: Example: For input 5, the pattern should be * *** ***** ******* ********* ******* ***** *** *	5	1,2	1,2,3
20.	Print the Pascal's triangle for a number of rows given by the user.	5	1,2	1,2,3